



of women opting to study this field is almost one-third of the total number of electronics engineers.

In order to encourage and foster an interest in STEM, we have come up with our own school connect program, called Prayas. Under this initiative we use an open source ARM-based hardware platform, called micro:bit, developed by BBC in the United Kingdom. PVP-SNDT-WU has collaborated with a few private schools and 50 schools run by Brihanmumbai Municipal Corporation (BMC).

We invite eighth and higher grade girl students to our campus during the summer vacations to build projects using the BBC micro:bit platform and develop technical thinking skills. We are doing this free of cost! We are trying to develop their interest in STEM and cultivate a curiosity towards electronics and computers. All members of our faculty participate in this exercise.

We conduct awareness programs in schools, reach out to the society by engaging with NGOs associated with us, involve our alumni to motivate our current college and school students. Fostering passion for electronics requires ongoing efforts and collaboration between educators, parents, role models, and the community.

What kind of jobs are available for polytechnic graduates and what does the job market look like currently?

We have an intake of 40 for the electronics polytechnic stream, out of which 10 to 12 students go for a job. The rest of the students go for higher studies. We're able to place all those students who are willing to work.

Our course is industry-integrated. We have around 25 companies associated with us, which are mainly SMEs. They always need

trained people and we don't find any difficulty placing our students.

Recruiters in SMEs have always said that they have a lot of vacancies, but they need competent candidates. Unfortunately, they do not get such candidates. If students are trained well at the diploma level, there are ample opportunities available in the industries.

While there are ample jobs in SMEs, if we talk about the opportunities in multinational companies, they are relatively lesser. On the bright side, there are plenty of startups and SMEs which are operating in and around Mumbai. We have also seen the job market evolve after the introduction of all of Make-in-India policies and startup culture after 2014.

Does the dearth of good candidates create a gap between industry and academia? How is your institution contributing towards the solution to this issue?

The industry-academia gap exists because industries expect freshers to have basic industrial skills and a certain level of capability which is not met by the graduates. If the capabilities are met, the gap is taken care of.

To counter that at SNDTWU, we have changed our curriculum to project-based learning. We use this methodology of project-based learning. Students should know basic things such as where to use a particular component, why to use a particular component, or how to read data sheets.

Even the coursework for all the subjects and the minor/major projects is designed to make the students solve problems that prepare them for the industry. For their final year projects, we make the students interact with industry experts to get valuable input. This is how industry and academia go hand-in-hand.

What kind of skills do you impart to the students to make them industry-ready?

Our students are well-versed in coding. Earlier, students would all get placed in IT companies because hardly any core company would come for recruitment to the campus.

With our focus on system-level programming things have changed. We start with a single-board computer such as an Arduino or Raspberry in the second year. If students want to later on migrate to IT they can do so as their programming fundamentals and basic computational thinking abilities are developed.

So, they have multiple domains to venture into—hardware and software in both embedded and IoT. They can move on to application programming if the opportunities are available.

What programming languages do you focus on and why?

Since it's an electronics course, we focus on hardware as well as software programming. Earlier, the focus was only on software programming languages such as C, which required a compiler on the PC. We focus on embedded C, which doesn't use the C compiler on the PC, but instead, we use a microcontroller such as 8051 or an Arduino system.

To teach Python we use Raspberry Pi as the hardware platform. Students learn how to interface sensors, process the data, and put it in a cloud. So now, when you learn programming in this way, you're learning how to program the hardware using the software, which is what electronics students need to study.

Do you believe that we have enough academicians who can aid in creating skilled graduates?

Yes, we do have enough academicians, but these people should be given opportunities. We have a very